Symbolic Music Generation with REMI Transformers

DeepMIR 2026 Fall — Homework 2

Chi-Jen Peng (R14725022)

Code, README, checkpoints, generated samples: **TODO:** cloud link

Repository: `https://github.com/chijenpeng/26-fall-DeepMIR-HW2`

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Experimental Setting

Experimental setting

Tasks

- Basic A: unconditional 32-bar generation, 3 inference settings $\times$ 20 pieces
- Basic B: 24-bar continuation of 3 TA prompts (8 bars), 3 settings
- Bonus (Option B): piano accompaniment for 10 melody prompts (16 bars)
- Data: all 896 POP909 training songs, no other data; models trained from scratch

Hardware / software

GPU	2 $\times$ RTX 4070 SUPER (12 GB), 1 $\times$ RTX A4500 (20 GB)
OS	Ubuntu 26.04
Software	PyTorch 2.8, Transformers 4.57.1, pretty_midi 0.2.11, FluidSynth

Method

Token representation: REMI on the annotated beat grid

- Bars start at the downbeats in `beat_midi.txt`; each bar: `Bar Beat_n Tempo_t`
- Events: `Pos_p` (16th note, 4 per beat) `[Chord_root:quality] Pitch Vel Dur`
- Vocabulary: 430 tokens; 32 bars: median 2,431 tokens, p99 3,900
- **TODO:** chords, velocity / duration bins, Basic vs. Bonus tracks

Models and training

TODO: GPT-2 (Basic) and T5 (Bonus) architecture, hyper-parameters, augmentation

Training length: held-out loss

TODO: held-out loss vs. step (13 prompt songs), choice of checkpoints

Basic Task A: Unconditional Generation

Basic A: inference settings and objective metrics

Model	Repr.	w/ Chord	Top-K / Top-P	Temp.	H1	H4	GS
GPT-2	REMI	O	K = 20	0.9	–	–	–
GPT-2	REMI	O	P = 0.9	1.0	–	–	–
GPT-2	REMI	O	K = 50	1.2	–	–	–
GPT-2	REMI	X	K = 20	0.9	–	–	–
Real data (896 songs)					2.123	2.626	0.877

20 pieces × 32 bars per row; metrics from the TA `eval_basic.py` (MusDr).

Basic Task B: Prompt Continuation

Basic B: continuation of the three prompts

TODO: settings, best setting per prompt (listening)

Bonus: Accompaniment Generation

Bonus: melody $\rightarrow$ piano with T5

TODO: model, inference, CA / OA table

Discussion

Discussion and limitations

TODO: findings

References

References

- **POP909**: Z. Wang et al., “POP909: A Pop-song Dataset for Music Arrangement Generation,” ISMIR 2020
- **REMI**: Y.-S. Huang, Y.-H. Yang, “Pop Music Transformer,” ACM MM 2020
- **MusDr** metrics: S.-L. Wu, Y.-H. Yang, “The Jazz Transformer on the Front Line,” ISMIR 2020
- **Bonus metrics**: Y. Ren et al., “PopMAG: Pop Music Accompaniment Generation,” ACM MM 2020
- **GPT-2**: A. Radford et al., 2019. **T5**: C. Raffel et al., JMLR 2020
- Hugging Face Transformers: T. Wolf et al., EMNLP 2020. PyTorch: A. Paszke et al., NeurIPS 2019
- TA scripts (`tokenization.py`, `eval_basic.py`, `eval_bonus.py`); TimGM6mb SoundFont (GPL-2)

Tool: Claude Code (Anthropic), OpenAI Codex.